



Anuj Lalla

Research Intern

Microsoft Research India

B.Tech., AI & Data Science (Graduated 2026)

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GitHub

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RESEARCH INTERESTS

Computer Vision Object Detection Vision-Language Models VLM Debiasing Multimodal Learning Generative Models
Agentic AI Mechanistic Interpretability Graph Representation Learning Domain Adaptation Domain Generalization
Continual Learning Machine Unlearning

HIGHLIGHTS

- **Branch Topper and Institute Silver Medalist**, B.Tech. AI & DS, IIT Jodhpur; **CGPA 9.05/10**.
- **1 paper accepted at CVPR 2026**; **1 paper under review at ICLR 2027**.
- **Research Intern, Microsoft Research India**, PowerPoint Research Group: **agentic and multimodal AI**.
- **4 papers reviewed**: CVPR 2026 workshop, CODS-COMAD 2026, IEEE JBHI; **many more subreviewed**.
- **TA at IIT Jodhpur and NPTEL**; supported a course with **nearly 8,000 enrollments**.

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (AI&DS)	Indian Institute of Technology, Jodhpur	9.05 (Department Rank : 1)	2022-2026
Higher Secondary	MSBSHSE (Maharashtra Board)	96.0 % (College Rank : 1)	2022
Secondary	MSBSHSE (Maharashtra Board)	95.8% (School Rank : 5)	2020

EXPERIENCE

- **Research Intern, Microsoft Research India** July 2026 – Present
PowerPoint Research Group
Manager: Dr. Vineeth N. Balasubramanian
 - Researched **agentic and multimodal AI** for automated PowerPoint generation and editing.
 - Investigated slide understanding, visual-layout reasoning, grounded content creation, and automated evaluation.
- **Course Teaching Assistant, NPTEL** June 2026 – Present
Deep Learning for Computer Vision (Instructor: Dr. Vineeth N. Balasubramanian)
 - Answered student queries for the 2026 offering, which had nearly **8,000 enrollments**.
 - Created the **course code-demo website** for the 2026 offering, hosting hands-on notebooks with Google Colab integration.
- **Research Intern, IIT Hyderabad** May 2025 – July 2025
Mentor: Dr. Vineeth N. Balasubramanian (MSR India) Source-Free Domain Adaptive Object Detection (CVPR'26 submission)
 - SFOD project targeting *foreground-background confusion* and *noisy pseudo-labels*.
 - Designed **SPAR**: simple consistency loss to suppress background activations (zero inference cost).
 - Designed **IRPL**: imbalance-aware entropy-regularized objective robust to noisy labels.
 - Benchmarked on Sim10k/KITTI/Cityscapes/BDD100K; consistently **+2-4 mAP** over strong SFOD baselines.
- **Undergraduate Research Assistant, FADE Lab, IIT Jodhpur** August 2024 – April 2026
Mentor: Dr. Pratik Mazumder GitHub | Video | Report
 - Studied **machine unlearning** for *domain generalization*, targeting safe forgetting while preserving OOD performance.
 - Reproduced **EASE** (CVPR 2024) on benchmark datasets for *class-incremental learning*.
 - Extended EASE with **GPM** (ICLR 2021) and **SGP** (AAAI 2023).

PUBLICATIONS

- **Foundation Model Priors Enhance Object Focus in Feature Space for Source-Free Object Detection.**
Accepted to the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. [arXiv]
Sairam VCR, Rishabh Lalla, Aveen Dayal, Tejal Kulkarni, **Anuj Lalla**, Vineeth N. Balasubramanian, Muhammad Haris Khan.

- **Reviewer (Conferences):** Domain Generalization: Evolution, Breakthroughs, and Future Horizons (2nd Edition), CVPR 2026 Workshop (1 paper); CODS-COMAD 2026 (2 papers).
- **Reviewer (Journals):** IEEE Journal of Biomedical and Health Informatics (1 paper).
- **Subreviewing:** Assisted with reviews of multiple additional papers.
- **Teaching Assistant,** Deep Learning (CSL4020, UG), IIT Jodhpur, Spring 2026; instructor: Dr. Pratik Mazumder.

COURSE PROJECTS

- **MAP7020: Applied ML Lab** Jan.–Apr. 2026
Interpreting Deepfake Detection on SID-Set GitHub
 - Benchmarked frozen **CLIP, DINOv2 and SigLIP** for three-class deepfake detection on SID-Set; a CLIP linear probe achieved **90.0 % macro-F1** without backbone fine-tuning.
 - Investigated detection cues using **SHAP, Token-CAM and layer-wise probing**, identifying class-specific, globally distributed signals and testing robustness to image degradation.
 - Compared **five sparse-autoencoder variants** on SIDA-7B embeddings, using feature ablation and **causal steering** to analyse detector decisions.
 - Exposed a confidence-stratified failure mode: accuracy fell to **12.7 %** on the lowest-confidence 20 % of tampered evaluation images; mapped adversarial vulnerability across classes.
- **CSL7860: Foundation Models and Generative AI** Aug.–Nov. 2025
CLALM: Continual Multilingual Visual Question Answering GitHub
 - Developed a **continual multilingual VQA pipeline** using **Qwen3-VL-2B and LoRA adapters** on ALM-Bench across 12 languages.
 - Compared naive sequential fine-tuning and **online EWC** under clustered, interleaved and randomized language orders, evaluating performance on previously seen languages.
 - Implemented per-language and question-type evaluation using **accuracy, exact match and token-level F1**, with a **Gradio demonstration** for comparing trained adapters.
 - **Additional assignments:** built an **MCP-based multi-agent paper reviewer** with schema and citation checks, and implemented **SISA unlearning** with shard/slice retraining and membership-inference evaluation.
 - Examinations and generative-model studies: compared **MambaVision, ViT and ConvNeXt** under ImageNet-R shift; tested LLMs with noisy prompts and long-context retrieval, explored ControlNet/MusicGen generation, and designed **IllusionScore** for image evaluation.
- **AIL7490: Multimodal Learning with Vision and Language** Mar.–Apr. 2026
ControlNet Reproduction and Spatially Conditioned Image Generation GitHub
 - Reproduced **ControlNet** over frozen **Stable Diffusion 1.5**, implementing zero convolutions, a conditioning encoder, the trainable UNet encoder copy and skip-connection injection from scratch.
 - Trained on **Fill50K** and evaluated spatially conditioned text-to-image generation using **FID, CLIP Score and PSNR**.
 - Investigated encoder-freezing configurations and the onset of conditioning during training; presented the paper's architecture and reproduction findings.
- **CSL7530: Advanced Machine Learning** Jan.–Apr. 2026
Advanced ML Algorithms: Implementation and Analysis GitHub
 - Implemented regression and multiclass **gradient boosting in NumPy**, including pseudo-residuals, shrinkage and Newton leaf updates; analysed regularization and robustness to outliers.
 - Built **Gaussian-process regression and Laplace-approximate classification** using NumPy/SciPy; investigated kernels, marginal-likelihood model selection and predictive uncertainty.
 - Implemented **VAE and conditional VAE** models in PyTorch for MNIST generation; examined latent representations and the trade-off between reconstruction and regularization.
 - Implemented **actor-critic PPO with GAE** on CartPole; compared clipped and unclipped objectives across three seeds using return variability and KL diagnostics.
- **AIL7390: Deep Learning for Natural Language Processing** Jan.–Apr. 2026
Cross-Lingual Sentence Embeddings and Neural Machine Translation GitHub
 - Fine-tuned **BGE-M3, Qwen3-Embedding and Jina embeddings** for Sanskrit–English alignment using **symmetric InfoNCE contrastive learning**.
 - Compared **Matryoshka-style embedding truncation at 32/64/128 dimensions** with learned projection heads, analysing cross-lingual alignment versus dimensionality.
 - Compared **MarianMT and ByT5**, subword versus byte-level tokenization, and beam-search decoding using **BLEU, BERTScore and inference-efficiency evaluation**.

- AIL7370: Deep Learning for Computer Vision** Jan.–Apr. 2026
Vision Architectures, Parallelism and Experimentation GitHub
 – Implemented **ConvLSTM**, **Siamese networks**, **Transformers** and **ViT** for spatio-temporal classification, metric learning and image-to-sequence tasks.
 – Compared **Faster R-CNN** and **DETR** on PASCAL VOC and implemented **TransUNet** for segmentation.
 – Studied **model and GPU parallelism**; used **Weights & Biases** for experiment tracking and visualisation.
- CSL7870: Learning on Graphs and its Applications** Aug.–Nov. 2025
Graph Representation Learning and Message-Passing Analysis GitHub
 – Compared **GraphSAGE aggregators** and **multi-head GAT** on Cora, including autoencoder-based feature compression and **oversmoothing** analysis.
 – Constructed **Graph2Vec-style embeddings** using **Weisfeiler–Lehman refinements**, **anonymous walks** and **Doc2Vec** for molecular graph classification.
 – Analysed **message-passing receptive fields**, **community mixing** and **structural equivalence** using BFS trees and Jaccard similarity.
- CSL7360: Computer Vision** Apr. 2025
Pruning-Aware Domain Generalization on PACS GitHub | Report | Demo
 Studied the interaction between **domain generalization** and **network pruning** on PACS using **DomainBed**, comparing ERM, ERM++, IRM, EQRM, URM and a HoG+MLP baseline. Evaluated unstructured, structured, random and SNIP pruning at 10–70 % across held-out domains. Unstructured pruning sometimes improved generalization, whereas structured and random pruning often degraded performance; ERM Cartoon accuracy increased from **78.92 %** to **80.93 %** at 50 % unstructured pruning.
- CSL4020: Deep Learning** Mar.–Apr. 2025
Single-Stage vs. Two-Stage Object Detection on KITTI GitHub | Report | Demo/Colab
 Compared **single-stage** and **two-stage detectors** by fine-tuning nine pretrained models on KITTI for 10 epochs, covering Faster R-CNN, RetinaNet, FCOS, SSD and YOLOv5x. Evaluated detection and localisation using mAP at multiple IoU thresholds, LRP/oLRP and qualitative bounding-box comparisons. Faster R-CNN ResNet-50-FPN v2 achieved the best **mAP@[0.50:0.95] of 0.519** and **oLRP of 0.3228**; YOLOv5x led the single-stage models with **0.459 mAP**.
- CSL7320: Digital Image Analysis** Apr. 2025
Classical Licence-Plate Localisation GitHub | Report
 Reimplemented edge-projection, morphological filtering and LBP-template baselines, and developed a hybrid pipeline combining LBP/HOG features with LightGBM for licence-plate localisation. Evaluated on **433 images**; morphology with stroke-width filtering performed best, achieving **F1 = 0.192** and **mIoU = 0.699**.
- MAL3010: Maths for Big Data** Apr. 2025
Parameter-Efficient Fine-Tuning of Vision Transformers Code | Report
 Compared **LoRA**, **LoHa**, **LoKr** and **AdaLoRA** for adapting ViT-Base to Food-101. Examined how low-rank, Hadamard and Kronecker decompositions trade off classification accuracy and trainable parameter counts across different adapter ranks.
- CSL7370: Dependable AI** Feb.–Mar. 2025
Robustness and Interpretability in Extreme Classification Report | Poster
 Extended **GalaXC** with cluster-based neighbor gating and hop-level/path explanations. Under adversarial edge injection, InterGalaXC achieved **18.4 % P@1 versus 1.8 %** for GalaXC on a 1,000-sample Amazon-131K evaluation. Additional coursework covered **adversarial attacks and defenses for classical classifiers** and **AnnexML explanations using Integrated Gradients and LIME**.
- CSL7770: Speech Understanding** Feb.–Apr. 2025
Audio Watermarking Robustness and Speech/Audio Analysis Project | Assignments | Report | Poster
 Evaluated **AudioSeal** watermarking on English/Hindi speech and reproduced AudioMarkBench robustness experiments with Square and single-query ALIF attacks. On 200 evaluation clips, ALIF yielded **0.41 bit accuracy**; frequency-selective adversarial training further reduced it to 0.35, exposing limitations of the tested defense. Coursework also compared STFT windows for UrbanSound8K classification and analysed music spectra across four genres.
- CSL4030: Data Engineering** Sept.–Nov. 2024
Data Pipelines and Database-Backed Applications GitHub | Report | Video
 Built a gold-price prediction pipeline with ETL, MySQL, Elasticsearch and a Dockerized Streamlit app. Also developed a MySQL-backed travel-booking system with user/admin workflows.
- CSL2050: Pattern Recognition and Machine Learning** Apr. 2024
Face Identification on Labelled Faces in the Wild GitHub | Website | Video | Demo
 Compared CNN, LBP and HoG features with PCA and classical classifiers for **face identification on LFW**. Evaluated SVM, logistic regression and tree-based models, and deployed a Streamlit demonstration. [Report]

• Earlier Coursework Projects

2023–2024

- **CSL3090: Artificial Intelligence:** Search, propositional theorem proving and Bayesian networks.
- **Chatroom (CSL2090):** Socket programming, file sharing and multithreading. [Video]
- **Photon Correlation:** Vectorized two-qubit simulation with Streamlit visualization.
- **Querimania:** Hostel complaint tracking with Google Forms and Apps Script. [Website]

KEY COURSES TAKEN

Completed at IIT Jodhpur, 2022–2026. Superscripts A and A* indicate the corresponding transcript grades.

Computer Science		Mathematics	Others (Lab and Other Discipline Courses)
Advanced Machine Learning ^A	Multimodal Learning with Vision and Language ^A	Probability, Statistics, and Stochastic Processes ^A	Applied ML Lab ^A
Deep Learning for Natural Language Processing ^A	Deep Learning for Computer Vision ^A	Linear Algebra and Differential Equations ^A	Introduction to Bioengineering Theory ^A
Autonomous Systems ^A	Learning on Graphs and its Applications ^A	Real Analysis	Engineering Mechanics
Foundation Models and Generative AI	Data Visualization ^A	Mathematics for Computing	Environmental Science
Speech Understanding	Digital Image Analysis ^A	Maths for Big Data	Foundations of Quantum Information ^A
Dependable AI ^A	Computer Vision ^A	Mathematics for Economics	Signals and Systems ^A
Deep Learning ^A	Optimization for Machine Learning ^A	-	Human Machine Interaction ^A
Design and Analysis of Algorithms	Artificial Intelligence	-	Thermodynamics
Data Engineering	Introduction to Computer Science	-	Engineering Visualization
Principles of Computer Systems 2	Principles of Computer Systems 1 ^{A*}	-	Gandhian Thought ^A
Data Structures and Algorithms	Pattern Recognition and Machine Learning ^A	-	-

TECHNICAL SKILLS

- **Programming:** Python, BashScript, SQL, MATLAB, LaTeX
- **Tools & OS:** Docker, ELK stack, MySQL, Apache NiFi, Git, Jupyter Notebook, Google Colab, Linux, Windows, qiskit, Venus Simulator (RISCV)
- **Libraries/Frameworks:** Pandas, Numpy, scikit-learn, PyTorch
- **Software:** Wireshark, Logisim, Solidworks
- **Web Skills:** HTML/CSS

ACHIEVEMENTS

- **Branch Topper (Silver Medalist)** 2026
Graduated **first in the B.Tech. in Artificial Intelligence and Data Science program (DR 1)** at IIT Jodhpur and received the **Institute Silver Medal**. Held DR 1 from the **fifth semester through graduation**.
- **Secured A* Grade** Ranked 1st in a class of 57 in the course Principles of Computer Systems-1 offered by Prof. Dip Sankar Banerjee and secured A* (exceptional) grade. 2024
- **CVIT Summer School on AI, IIIT Hyderabad** [Certificate] 2025
Attended in July 2025, covering multimodal AI, transformer reasoning, vision-language and speech models, and medical AI, with hands-on practice.
- **Computational Intelligence and Generative AI Winter School** [Certificate] 2025
Attended at IIT Indore in January 2025, focusing on language-model adaptation and interpretability, empathetic AI, and AI for science in molecular biology and medicine.
- **Outstanding Academic Performance** 2022–2026
Earned **A grades in 23 courses**; selected A-grade courses are marked with superscript A in the **Key Courses Taken** table.
- **HSC Computer Science: 200/200** MSBSHSE [Marksheet] 2022
- **Quantum Computing, Qubit by Qubit** [Certificate] 2020–2021
Completed a two-semester course covering quantum algorithms and practical programming with Qiskit.